

The Race: Decomposing PipelineRL into Concurrency and In-Flight Weight Updates, and Measuring What Each One Actually Buys

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Abstract

Reinforcement learning on language models alternates between two workloads that want opposite hardware: *generation*, which is memory-bound and finishes at wildly different times per sequence, and *training*, which is compute-bound and synchronous. Conventional RL frameworks such as veRL run them in turn, so the cluster idles while it waits for the slowest sequence in each batch. PipelineRL (Piché et al., 2025) removes that idle time by running generation and training *concurrently* on disjoint GPUs and pushing new weights into the inference engine *mid-sequence*, without discarding the KV cache. The reported speedup is $\approx 2\times$ on 128 H100s. We observe that this headline bundles *two* independent mechanisms, and that no published measurement separates them. We therefore run three arms that share model, data, GRPO implementation, batch shape, seed, library versions and total GPU budget, and differ only in *when each GPU does what*: (A) conventional alternation; (B) concurrent generation and training with weights applied between generation batches; (C) concurrent with genuine in-flight updates. On $2\times$ H100 with Qwen2.5-1.5B on GSM8K, 45 minutes per arm, we find that **concurrency alone doubles throughput** ($14.3 \rightarrow 29.4$ samples/s) and cuts idle GPU-time $7.2\times$ ($1861 \rightarrow 258$ GPU-seconds), while **in-flight updates — which demonstrably work, 514/514 landing mid-decode with the cache retained — buy nothing measurable at this scale**, halving an already-negligible staleness ($0.051 \rightarrow 0.035$ steps) at a $\sim 7\%$ throughput cost. We also report a result we cannot yet attribute: both concurrent arms end lower on reward, and the arms begin at different rewards *before* training can separate them, which we trace to a tensor-parallel difference in the generator rather than to the scheduler. We release the harness, which enforces its own fairness contract in code — refusing to emit a figure when GPU-second accounting fails, when an arm leaves a GPU idle, or when any arm’s environment fingerprint differs.

1 Introduction

The RL loop for a language model is a systems problem before it is an algorithmic one. A policy generates completions, a verifier scores them, and a trainer takes a gradient step. The generation phase is memory-hungry and embarrassingly parallel; the training phase is compute-hungry and synchronous. Running both on the same accelerators means one of them is always waiting.

Conventional implementations accept this and alternate. veRL / HybridFlow [1] makes the alternation as cheap as possible — resharding the actor between an inference-optimal and a training-optimal layout with its 3D-HybridEngine, and searching for a good device placement automatically — and reports $1.5\text{--}20\times$ throughput over prior systems. But the alternation itself remains: while any sequence in the batch is still decoding, the GPUs that finished early have nothing to do.

PipelineRL [2] attacks that residual idle time directly. Its two changes are:

1. **Concurrency.** Partition the cluster. Some GPUs generate forever; others train forever. Neither waits for the other’s phase.
2. **In-flight weight updates.** Because the generators never stop, their weights would grow stale. So the trainer pushes new weights *into a running engine*: generation pauses between decode steps, the weights are swapped, the KV cache is retained, and sequences already in progress continue under the new policy — producing sequences whose early tokens came from an older policy than their late tokens.

These are logically independent. One could adopt concurrency alone and update weights at batch boundaries; one could not sensibly do the reverse. The published comparison is conventional versus *both*, which leaves an obvious question unanswered: of the reported speedup, how much is the partition and how much is the mid-sequence update?

Contributions.

- A three-arm decomposition (§2) isolating concurrency from in-flight updates, with a fairness contract enforced mechanically rather than by convention.
- A measurement (§4) showing concurrency accounts for essentially the entire throughput gain at small scale, and that in-flight updates are inert there — with an explanation of the regime in which they should matter.
- A negative result and an honest attribution failure (§5): the concurrent arms learn less per sample, but a generator-side confound prevents us from attributing that to staleness.
- An open-source harness (§6) whose validators refuse to publish malformed comparisons — built after several of our own runs produced confident, well-formed, wrong numbers.

2 Method

2.1 The three arms

All arms optimise the same objective with the same code. We use GRPO [3]: for each prompt, G completions are sampled, the group’s mean reward forms the baseline, and the token-level clipped policy-gradient loss is applied with group-normalised advantages. The reward is binary and verifiable — 1 if the final numeric answer matches GSM8K’s gold answer, else 0 — so there is no reward model and nothing to hack.

A — Conventional. All N GPUs alternate. vLLM runs tensor-parallel across all of them for generation; the engine then sleeps, releasing its KV cache so the trainer can use the memory; the trainer steps; weights are pushed back and the engine wakes. Data is always perfectly on-policy (lag = 0 by construction). This is the colocated HybridEngine pattern, and the PipelineRL authors note that veRL’s throughput should resemble their conventional baseline.

B — Concurrent, batch-boundary updates. The GPUs are partitioned once: k generate continuously, $N - k$ train continuously. A producer keeps the inference engine saturated; a consumer trains on completed rollouts. Weights are applied *between* generation batches.

C — Concurrent, in-flight updates. Identical partition, but the update is issued against a running engine via `pause_generation(mode="keep", clear_cache=False), collective_rpc_async, resume_generation`. The engine halts between decode steps, in-flight requests and their KV blocks survive, and decoding resumes under the new weights.

2.2 The fairness contract

A three-way comparison is worth nothing unless the arms differ in exactly one respect. We enforce this in code rather than by discipline:

- **One GRPO implementation and one batching path**, imported by all arms. No arm has its own loss.
- A `fairness_key()` covering every hyper-parameter *and an environment fingerprint* (vLLM / PyTorch / Transformers versions, GPU model), compared after the run. A mismatch aborts publication. This is not hypothetical: the in-flight arm requires $vLLM \geq 0.18$, and an earlier iteration of this study ran arms on 0.11 and 0.18, which would have confounded “scheduling” with “inference engine upgraded”.
- A **placement check**: each arm verifies at start-up that its trainer is not on an inference GPU and that every inference GPU actually holds an engine.
- A **GPU-second invariant**: every GPU is in exactly one state at every instant, so accounted GPU-seconds must equal $N \times \text{duration}$. Violation means the event stream is out of order, and the analyser refuses to report a utilisation split rather than reporting a plausible one.
- An **unused-GPU check**: an arm that leaves a GPU idle raced with a handicap. Our first four-GPU attempt did exactly this — the trainer was single-GPU while the partition allocated two — and the resulting “conventional wins” was an artefact of the missing GPU.

The only quantity an arm may interpret differently is how the fixed GPU budget is *partitioned*.

2.3 Telemetry

Every GPU state transition, weight transfer, and finished rollout appends one line to a single JSONL stream. All figures and all reported numbers derive from that one stream, so a plot and an animation of the same run cannot disagree. Idle is emitted explicitly and eagerly; a stream that omitted idleness would make the arms look identical. Timestamps are taken under the writer’s lock — sampling them outside it allowed two threads to stamp out-of-order lines, which inflated integrated GPU-seconds beyond the cluster’s capacity and was caught only by the invariant above.

3 Experimental setup

Qwen2.5-1.5B-Instruct on GSM8K, $2 \times H100$, 45 minutes of wall-clock per arm after engine start-up (reported separately). $G = 8$ completions per prompt, 16 prompts per optimizer step (128 samples/step), up to 512 new tokens, temperature 1.0, learning rate 10^{-6} , clipping $\varepsilon = 0.2$, no KL penalty. Arms B and C use one generation GPU and one training GPU; arm A time-shares both.

Two GPUs rather than four is a deliberate consequence of the fairness contract. Our trainer is single-GPU, so a fair split needs $k = N - 1$ generation GPUs; at $N = 4$ that requires $TP = 3$, which vLLM cannot shard for a model with 2 key-value heads. Two GPUs is therefore the largest configuration in which no GPU is stranded. We regard this as a real limitation of the study, not a detail (§5).

Weight transfer uses CUDA IPC handles: `reduce_tensor` yields a picklable handle the inference workers reopen against the sender’s GPU memory, so nothing crosses PCIe. The transport is chosen *before* the first push from `torch.cuda.can_device_access_peer`, never by catching an exception — an exception inside `collective_rpc` poisons the vLLM executor, so a runtime fallback would kill the engine on first use.

4 Results

	A conventional	B concurrent	C concurrent+in-flight
samples generated	38,784	80,256	74,112
samples / second	14.34	29.37	27.39
vs. conventional	—	2.05×	1.91×
GPU-seconds idle	1861.1	1638.4	257.7
GPUs computing (mean)	65.6%	70.0%	95.2%
optimizer steps	303	562	514
weight updates	303	562	514
of which in-flight	0	0	514
mean weight-push (ms)	453.3	319.9	260.5
mean lag (optimizer steps)	0.0	0.051	0.035
reward, final 10% of samples	0.7476	0.6888	0.6826
reward at matched samples	0.7476	0.6978	0.6929

Table 1: All three arms, identical budget and hyper-parameters, 45 minutes each. The fairness key — including library versions — matched exactly across arms.

4.1 Concurrency doubles throughput

Arm B generates $2.05\times$ the samples per second of arm A. Figure 1 shows the mechanism directly: the fraction of GPUs actually computing, over time. Arm A sags on every generate/train alternation and averages 64%; arm C is a near-solid block at 95%. The white space above each curve is compute that was paid for and not used, and it falls $7.2\times$ from arm A to arm C.

4.2 In-flight updates work — and are inert at this scale

All 514 of arm C’s weight updates landed mid-decode with the KV cache retained; we verified separately on a live engine that the KV block count is unchanged across a pause–update–resume cycle while a 384-token generation is in flight, and that the request still completes. The mechanism is real.

Its effect is not. Arm C halves mean lag relative to arm B ($0.051 \rightarrow 0.035$ optimizer steps) and costs $\sim 7\%$ throughput, and Figure 2(right) shows the two concurrent arms’ learning curves lying on top of one another. The explanation is simply that at two GPUs there is almost no staleness to remove: 95% of arm B’s samples are *already* perfectly on-policy, because a generation batch completes in roughly the time of one optimizer step. In-flight updates are a solution to a problem that does not exist at this scale.

We would expect them to matter when lag is large — many generation GPUs per trainer, long sequences, high length variance — which is precisely the 128-GPU regime the original paper measures.

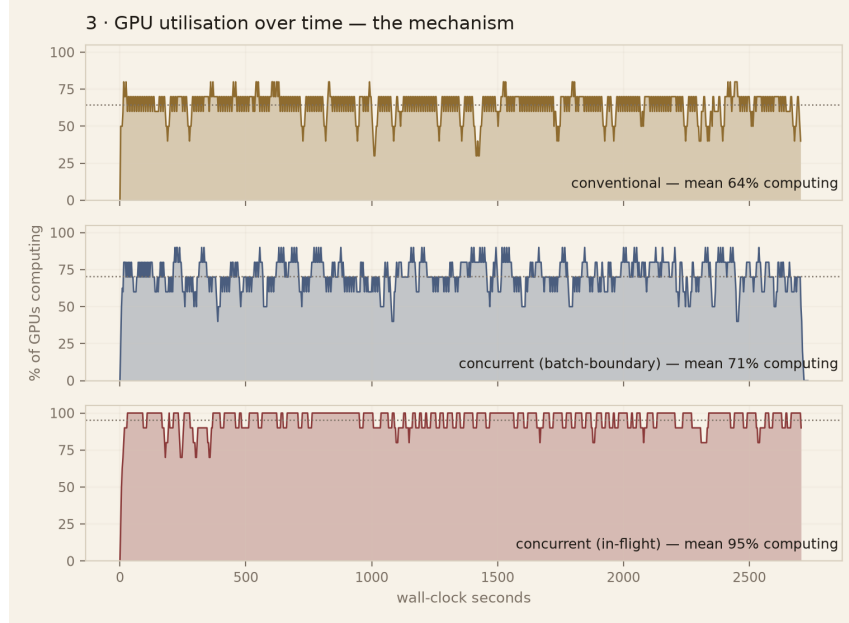


Figure 1: GPU utilisation over time. The alternation bubble is visible in arm A as periodic sags; the in-flight arm removes it almost entirely.

4.3 Small staleness, visible in the loss

Although lag is small, it is not invisible. Figure 3 shows that the fraction of tokens whose importance ratio leaves the trust region climbs from $\approx 0.12\%$ to $\approx 0.35\%$ over training for both concurrent arms, while the conventional arm stays flat. As the policy moves faster, the same one-step lag displaces the ratio further. This is the mechanism by which staleness would eventually cost effectiveness — observable here, but far too small to explain the reward gap in Table 1.

5 Limitations and an unattributed result

The concurrent arms learn less per sample, and we cannot say why. At matched sample budget — identical prompts in identical order — arm A reaches 0.7476 against 0.6978 and 0.6929. That is a real difference, but we decline to attribute it to staleness, because the arms *begin* at different rewards: over their first 10% of samples, 0.6650, 0.6510 and 0.6417 respectively, before training could have separated them. Same base model, same seed, same prompts.

The likely cause is the generator, not the scheduler. Arm A generates with $TP = 2$ and arms B and C with $TP = 1$; tensor-parallel reduction order changes logits at the last bit, and under temperature-1 sampling a perturbed logit changes the entire sampled trajectory. The arms are therefore drawing from measurably different sampling distributions before any learning occurs. Settling this requires (i) evaluating on a held-out set with fixed decoding rather than reading reward off training rollouts, and (ii) matching generator tensor-parallelism across arms, e.g. by running data-parallel $TP = 1$ engines in the conventional arm. We regard the throughput and utilisation results as sound and the learning comparison as open.

Scale. Two GPUs compresses the effect under study. The idle bubble grows with cluster size and with sequence-length variance; a two-GPU cluster has one generation GPU, so there is no “other GPUs wait for the straggler” effect at all — only the phase alternation. Our headline is therefore a lower bound on what

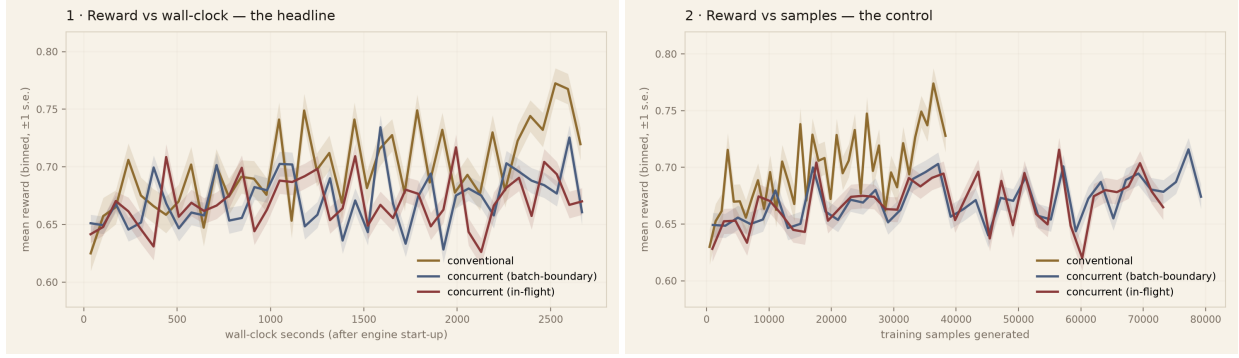


Figure 2: Reward against wall-clock (left) and against samples generated (right). The two concurrent arms coincide, indicating in-flight updates changed learning by nothing measurable. Both sit below conventional — see §5 for why we do not attribute that to the scheduler.

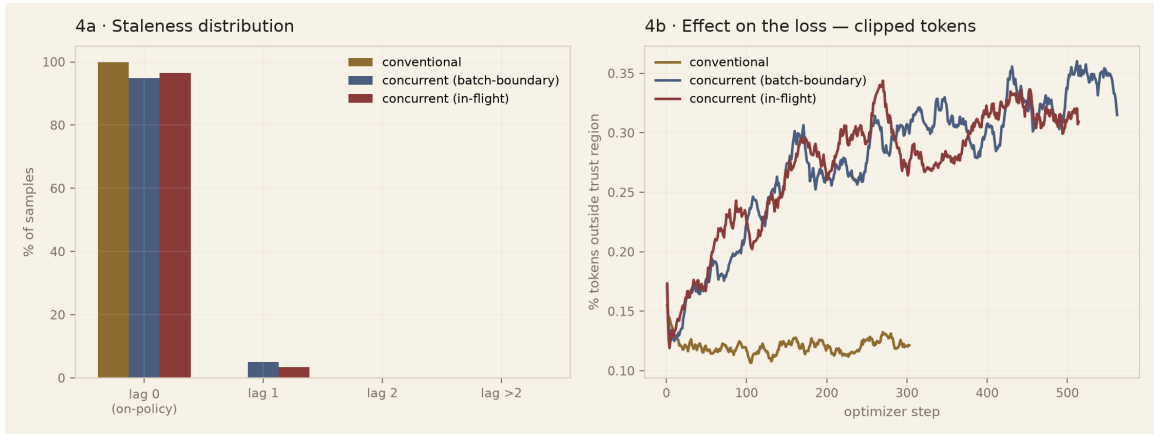


Figure 3: Left: distribution of per-sample lag. Right: clipped-token fraction over training, the effect of that lag on the loss.

concurrency buys.

Staleness measurement. Lag is recorded when a sequence *completes*, which is a lower bound: a sequence spanning an in-flight update has tokens of several policy ages, and we record only the count of versions that landed while it decoded. Reporting per-token effective sample size would be stricter; our harness now logs the necessary statistic, but the runs reported here predate it and we do not substitute a proxy silently.

Single seed. Each arm was run once. The throughput and utilisation differences are far outside plausible run-to-run variation; the reward differences are not obviously so.

6 The harness

Building this taught us more than the result did. Six distinct bugs in our own code produced *confident, well-formed, wrong* output rather than crashing: telemetry timestamps taken outside the writer’s lock; a vLLM executor that returns a stale error to every subsequent call after one failure, making a healthy component report as broken; a `CUDA_VISIBLE_DEVICES` leak that silently co-located the trainer with the generator while still reporting a clean split; an optimizer-offload ordering error; a utilisation metric counting waiting

as busy, which reported 97.8% for an arm with a completely idle GPU; and a GRPO grouping error in which one rollout per prompt made the advantage baseline average over unrelated questions.

Every one of these is now an assertion that refuses to proceed, and the assertions are the part of this artifact we would most encourage others to copy. A comparison harness that cannot fail loudly will eventually publish something false.

7 Related work

veRL / HybridFlow [1] formalises RLHF as a hybrid-controller dataflow and makes the colocated alternation efficient. PipelineRL [2] introduces concurrent generation/training with in-flight weight updates. GRPO [3] removes the critic by using a group baseline. Our contribution is not a new method but a decomposition and an honest measurement of an existing one at small scale, plus the validation harness that makes such measurements trustworthy.

References

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